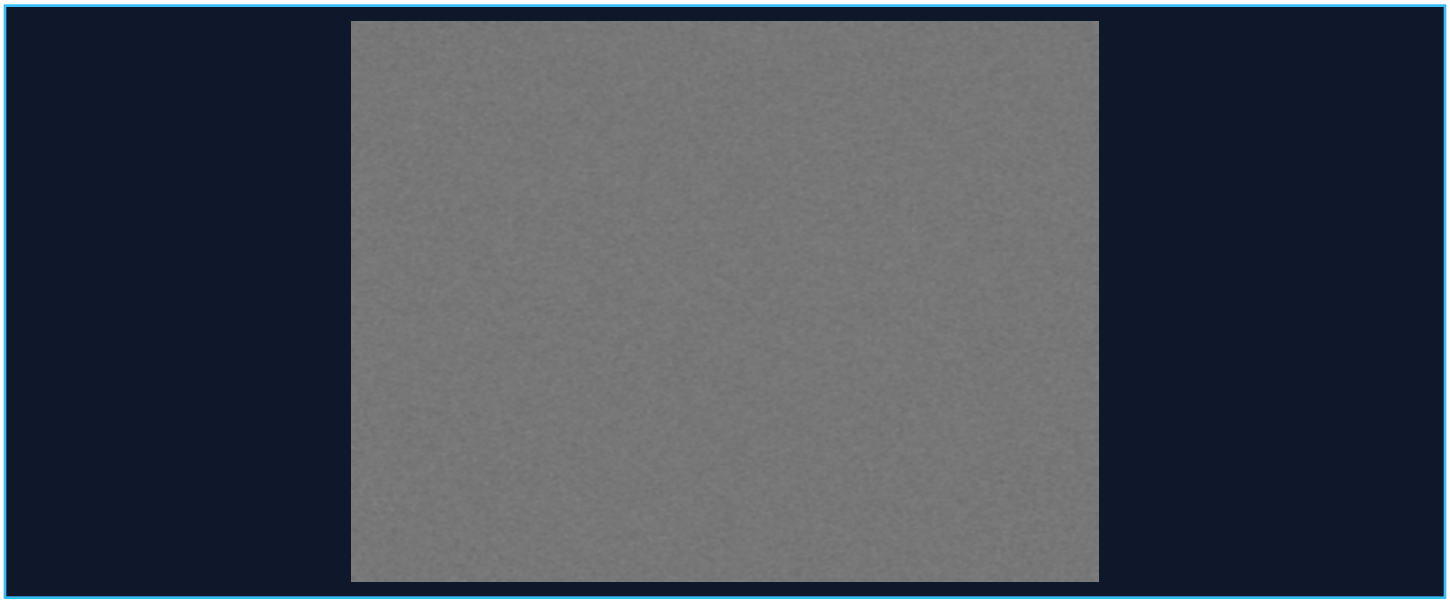
ASTROSIGHT | PLANETARY SURFACE INTELLIGENCE

AUTONOMOUS LUNAR & MARTIAN TOPOGRAPHICAL CRATER ANALYSIS REPORT

Analyst / Researcher:	Dr. Alex Reynolds	**Target Body:**	**MOON**
Timestamp:	250d7a7b-5912-4df2-	**Resolution (GSD):**	10.0 m/px
Validation Status:	**VALID_NO_CRATER**	**Craters Identified:**	0
Model Pipeline:	AstroSight-CraterNet-v2.4	**Processing Time:**	63 ms

1. TOPOGRAPHICAL DETECTION VISUALIZATION



2. CRATER FEATURE COORDINATES (X, Y)

No confident crater features were detected on this planetary terrain specimen.

3. EUCLIDEAN & REAL-WORLD SPATIAL DISTANCE MATRIX

No spatial distance vectors calculated for this analysis run.

4. SCIENTIFIC SPECIFICATIONS & UN SDG 9

Mathematical Formulation: Planar Euclidean metric d = √((x■ - x■)² + (y■ - y■)²) scaled via orthorectified ground sampling distance 10.0 m/px. Real-world distance D = d × resolution.
United Nations SDG 9 Alignment: AstroSight advances UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by establishing open AI infrastructure for planetary remote sensing, crater topology, and deep-space mission research.

Scientific Disclaimer: *AstroSight automated crater boundary coordinates and distance vectors are generated for scientific research and mission planning. Topographical scale assumes standard planar projection for small-area orbital imagery. Always cross-correlate with high-resolution LROC NAC / HiRISE elevation models.*